1. 当 $n=0$

有 $6 \mid 0(0+1)(2\times0+1)$

当 $6 \mid k(k+1)(2k+1)$，证明 $k=k+1$ 时也成立.

$S=(k+1)(k+1)(k+1)(2k+1)$

$= 2k^3+9k^2+13k+6$

$= k(k+1)(2k+1)+6(k+2)(k+1)$

$\because 6 \mid k(k+1)(2k+1)$

$\therefore 6 \mid S$

即命题对任意成立

$\therefore 6 \mid n(n+1)(2n+1)$

2. n 个连续整数可表示为 $a, a+1, \cdots, a+n-1$

对于 a 和 n，存在唯一整数 q 和 r $(0 \le r < n)$

使 $a = nq+r$

则表示为 $nq+r, nq+r+1, \cdots, nq+r+n-1$

其中 $nq+r+(n-r) = n(q+1)$ 可被 n 除尽

且这个数是唯一的

3. $\because m-p \mid mn+qp$

$\therefore mn+qp = k(m-p)$

$mq+np = (mn+qp) - n(m-p) + p(m-p)$

$= (k-n+p)(m-p)$

$\therefore m-p \mid mq+np$.

4. $d=(a+b, a-b) \Rightarrow d \mid a+b \quad d \mid a-b$

$\therefore d \mid [(a+b)+(a-b)]$ 且 $d \mid [(a+b)-(a-b)]$

$\therefore d \mid 2a$ 且 $d \mid 2b$.

又 $(a,b)=1$

$\therefore d \mid 2$

$\therefore d = 2$ 或 $1 \Rightarrow (a+b, a-b) = 1$ 或 2

5. $\because (a,b)=1 \quad (c,d)=1$

$\frac{a}{b}+\frac{c}{d} = N = \frac{ad+bc}{bd}$

$ad+bc = Nbd$.

$ad = Nbd-bc$

$= b(Nd-c)$

$\therefore b \mid ad$

$\because (b,a)=1$

$\therefore b \mid d$

同理 $d \mid b$

$\therefore |b| = |d|$

6. 设 $d=(a,b)$

$a = d a_1$

$b = d b_1$

$(a_1, b_1)=1$

$(am, bm) = (d a_1 m, d b_1 m)$

$= d \cdot (a_1 m, b_1 m)$

$\because (a_1, b_1)=1$

$\therefore (a_1 m, b_1 m) = m$

$\therefore (am, bm) = d \cdot m = (a \cdot b) m$.

7. 即证 $(14n+3, 21n+4)=1$

令 $(14n+3, 21n+4)=d$

$d \mid 14n+3, \quad d \mid 21n+4$

$\therefore d \mid 3\times(14n+3) - 2\times(21n+4)$

$\Rightarrow d \mid 1$

$\therefore d=1$

8. 当 $n=1$，$P_1=2$，$2^{2^1}=4$，$2<4$，成立

若当 $n=k$ 时成立，$P_k < 2^{2^k}$，考虑 $n=k+1$

令 $M = P_1 P_2 \cdots P_k + 1$

设 P_{k+1} 是 M 的素因数

$\because (M, P_j)=1$ 且 P_j 包含先前所有素数

$\therefore P_{k+1} > P_k$

$\because P_j < 2^{2^j}$

$\therefore M < (2^{2^k})^2 = 2^{2^{k+1}}$

$\therefore P_{k+1} < 2^{2^{k+1}}$

$\therefore P_n < 2^{2^n}$.

9. (1) $1245 \div 233 = 5 \cdots 80$

$233 \div 80 = 2 \cdots 73$

$80 \div 73 = 1 \cdots 7$

$73 \div 7 = 10 \cdots 3$

$7 \div 3 = 2 \cdots 1$

$3 \div 1 = 3 \cdots 0$

$\therefore \gcd(1245, 233) = 1$

(2) $211 \div 189 = 1 \cdots 22$.

$189 \div 22 = 8 \cdots 13$

$22 \div 13 = 1 \cdots 9$

$13 \div 9 = 1 \cdots 4$

$9 \div 4 = 2 \cdots 1$

$4 \div 1 = 4 \cdots 0$

$\therefore \gcd(189, 211) = 1$

10. 求 x, y 使 $1895x + 782y = 1$

$1895 \div 782 = 2 \cdots 331$

$782 \div 331 = 2 \cdots 120$

$331 \div 120 = 2 \cdots 91$

$120 \div 91 = 1 \cdots 29$

$91 \div 29 = 3 \cdots 4$

$29 \div 4 = 7 \cdots 1$

代回：得 $1 = 29 - 4\times7$

$= 29 - 7\times(91-3\times29) = 22\times29 - 7\times91$

$= 22\times(120-91) - 7\times91 = 22\times120 - 29\times91$

$= 22\times120 - 29\times(331-2\times120) = 80\times120 - 29\times331$

$= 80\times(782-2\times331) - 29\times331 = 80\times782 - 189\times331$

$= 80\times782 - 189\times(1895-2\times782)$

$= -189\times1895 + 458\times782$

$\therefore 782^{-1} \bmod 1895 = 458$